

NGUYEN GIA HUY

Cloud Engineer

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PROFESSIONAL SUMMARY

Cloud engineer with nearly three years of experience across private and public cloud. Designed and operated VNPAY Cloud's **OpenStack**-based private cloud hosting **50+ production Kubernetes clusters** for fintech and enterprise customers, with Keycloak-based tenant isolation for **20+ customer organizations** and Kong Gateway access control for **100+ services**. On **AWS**, deployed Thought Machine Vault Core on EKS, RDS, ECR, and Elastic Load Balancing for a digital bank, ran account migration on ECS and Lambda, and built serverless and containerized data pipelines on Lambda, ECS, S3, Athena, and RDS for a utility platform serving millions of users.

TECHNICAL SKILLS

Private Cloud:	OpenStack, Cluster API, Kubernetes
AWS Compute & Containers:	Amazon EKS, Amazon ECS, AWS Lambda, Amazon ECR
AWS Storage, Data & Network:	Amazon RDS, Amazon S3, Amazon Athena, Elastic Load Balancing
Provisioning:	Terraform, AWS SAM, Cluster API
Identity & Security:	Keycloak, RBAC, multi-tenant isolation, HashiCorp Vault
Service Networking:	Kong Gateway (service- and network-level access control), Consul
Cloud Operations:	CloudWatch, DataDog, Prometheus, Grafana, OpenTelemetry
Languages:	Golang, Python

WORK EXPERIENCE

Middle Software Engineer – AWS Cloud Infrastructure 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Deployed a new **Thought Machine Vault Core** version on AWS for Vikki Digital Bank's core banking platform.
- Ran the platform on **Amazon EKS**, with **Amazon RDS** as its managed database, **Amazon ECR** as its container registry, and **Elastic Load Balancing** in front of it.
- Automated releases of the platform through a CI/CD pipeline covering build, test, and deployment.
- Built the account migration system on a mix of **ECS** containers and **Lambda** functions over RDS (Golang), reducing migration time from **2.5 hours to 27 minutes (5x faster)**.
- Integrated the cloud-hosted Vault Core with the bank's external backend services for core banking operations and transaction workflows.
- Cut simulation test time for the platform from **24 minutes to just over 50 seconds (20x)** by rebuilding the Gauge-based framework in Golang.
- On the application side, developed and maintained Lending and Overdraft products (Python Smart Contracts): interest accrual, penalty and late fees, repayment, and settlement.

Software Engineer – AWS Data Platform 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Designed and operated serverless and containerized **ETL pipelines** on **AWS Lambda** and **Amazon ECS** for au Denki's large-scale utility data.
- Organized storage and querying on **Amazon S3** and **Amazon Athena** for analytics, with **Amazon RDS** serving relational and operational data.
- Provisioned the platform's AWS resources with **Terraform** and its serverless components with **AWS SAM**.
- Monitored the AWS workloads with **CloudWatch** and **DataDog** for pipeline health, failures, and operational anomalies.
- Scaled batch processing throughput by **150%** through pipeline redesign, parallel execution, and more efficient data access.
- Migrated search from **Elasticsearch to OpenSearch**, redesigning ingestion and indexing flows for stability, consistency, and query performance.

Cloud Engineer, OpenStack & Kubernetes (Software Engineer) 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

Private Cloud Architecture

- Designed the private cloud platform behind VNPAY Cloud on **OpenStack** infrastructure, serving a fintech and enterprise customer base.
- Operated the platform in production at a scale of **50+ Kubernetes clusters**.
- Used **Cluster API** to provision Kubernetes clusters on OpenStack declaratively, making new cluster capacity an automated operation.

- Scaled clusters through Cluster API-driven automation instead of hand-built infrastructure work.
- Upgraded production clusters with **zero downtime** through Cluster API's rolling machine replacement.

Multi-tenant Identity and Access

- Implemented tenant isolation for **20+ customer organizations** on shared cloud infrastructure, backed by **Keycloak** multi-tenant authentication.
- Implemented role-based access control so that each tenant's users only reach their own organization's resources.

Service Networking and Automation

- Integrated and operated **Kong Gateway** as the access layer for a microservices ecosystem of **100+ services**.
- Controlled access at both the service level and the network level through gateway policies.
- Automated about **80%** of data center provisioning and infrastructure configuration, cutting manual effort in infrastructure delivery.

Blockchain Developer

06/2023 – 09/2023

EsoiLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built event-driven services bridging on-chain events and off-chain databases and APIs.
- Connected the services to **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui), ensuring consistent transaction handling and data integrity.
- Developed and reviewed **15+ smart contracts** in Solidity (EVM chains) and Move (Aptos, Sui) for DeFi and NFT use cases.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Ran the backend across multiple clusters with **failover**, monitored through OpenTelemetry, Prometheus, Grafana, ELK, and ClickHouse, at **99.95% uptime**.
- Built the backend on gRPC-Gateway, Kafka, MongoDB, and Redis, serving up to **1.5K requests/second** at **80–100 ms** latency.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on LiveKit with OpenAI models for live transcription, voice translation, and summarization, and designed its real-time translation pipeline.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera